NET732

Week10:Incident Response

Practical

1. Creating a Week 10 folder

I created a folder to store my Week 10 practical work and evidence.

mkdir -p ~/CyberOps/Week10

cd ~/CyberOps/Week10

2. Simulating a security incident

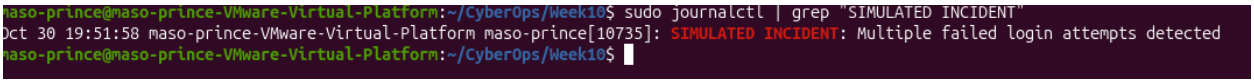
I used the following command to create a simulated security warning in the system logs.

logger -p auth.warning "SIMULATED INCIDENT: Multiple failed login attempts detected"

I then checked the logs to find the simulated incident.

sudo journalctl -n 50

sudo journalctl | grep "SIMULATED INCIDENT"



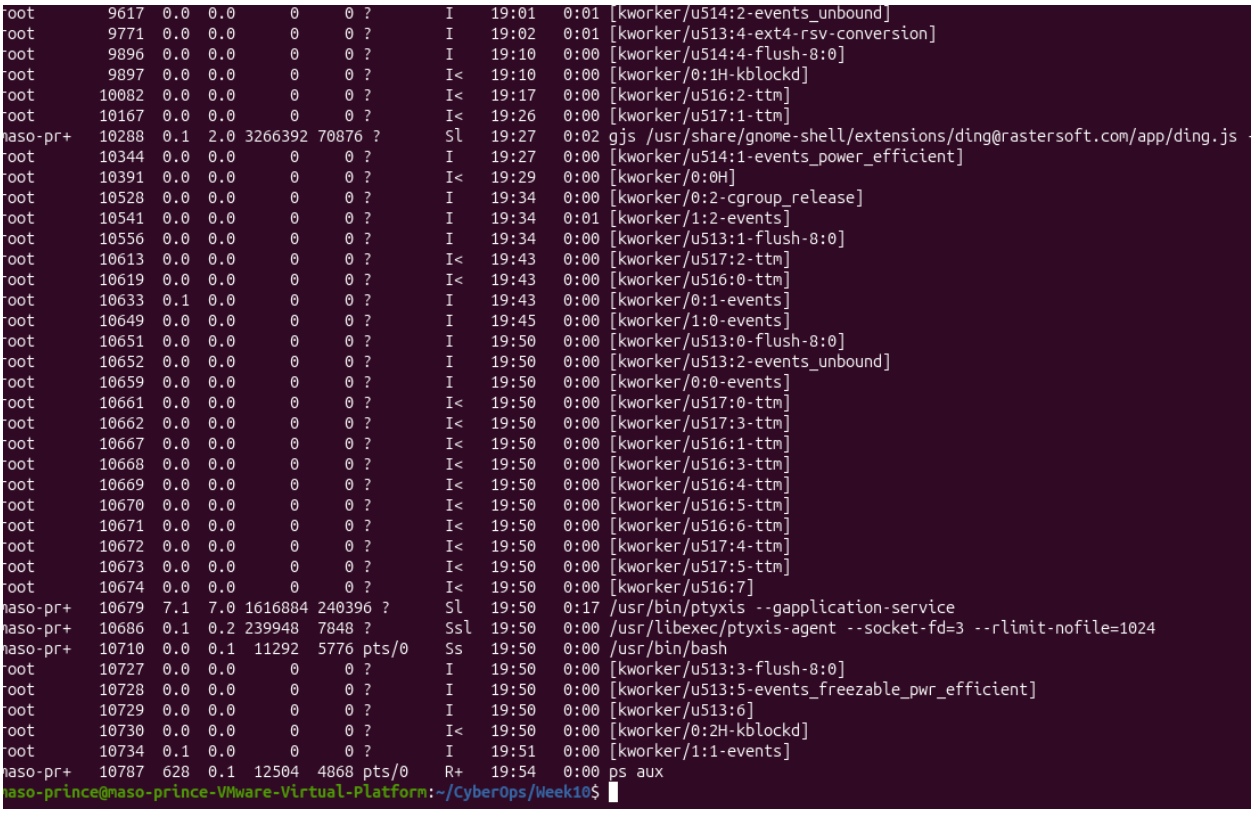
3. Investigating the incident

I used these commands to check login activity, failed login attempts and running processes.

last

sudo journalctl | grep -iE "failed|invalid user|authentication failure"

ps aux



This helped me understand how to investigate suspicious activities on a system.

4. Containment

I planned to isolate the affected machine from the network to prevent further suspicious activity.

First, I checked my network interface:

ip addr

I planned to temporarily disable the network interface using:

sudo ip link set ens33 down

I would replace ens33 with my actual interface name.

```
root       10669  0.0  0.0      0     0 ?        I<   19:50   0:00 [kworker/u516:4-ttm]
root       10670  0.0  0.0      0     0 ?        I<   19:50   0:00 [kworker/u516:5-ttm]
root       10671  0.0  0.0      0     0 ?        I<   19:50   0:00 [kworker/u516:6-ttm]
root       10672  0.0  0.0      0     0 ?        I<   19:50   0:00 [kworker/u517:4-ttm]
root       10673  0.0  0.0      0     0 ?        I<   19:50   0:00 [kworker/u517:5-ttm]
root       10674  0.0  0.0      0     0 ?        I<   19:50   0:00 [kworker/u516:7]
maso-pr+   10679  7.1  7.0 1616884 240396 ?      Sl   19:50   0:17 /usr/bin/ptyxis --gapplication-service
maso-pr+   10686  0.1  0.2 239948  7848 ?        Ssl  19:50   0:00 /usr/libexec/ptyxis-agent --socket-fd=3 --rlimit-nofile=1024
maso-pr+   10710  0.0  0.1  11292  5776 pts/0    Ss   19:50   0:00 /usr/bin/bash
root       10727  0.0  0.0      0     0 ?        I    19:50   0:00 [kworker/u513:3-flush-8:0]
root       10728  0.0  0.0      0     0 ?        I    19:50   0:00 [kworker/u513:5-events_freezable_pwr_efficient]
root       10729  0.0  0.0      0     0 ?        I    19:50   0:00 [kworker/u513:6]
root       10730  0.0  0.0      0     0 ?        I<   19:50   0:00 [kworker/0:2H-kblockd]
root       10734  0.1  0.0      0     0 ?        I    19:51   0:00 [kworker/1:1-events]
maso-pr+   10787  628  0.1  12504  4868 pts/0    R+   19:54   0:00 ps aux
maso-prince@maso-prince-VMware-Virtual-Platform:~/CyberOps/Week10$ ip addr
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
       valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
       valid_lft forever preferred_lft forever
2: ens33: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UNKNOWN group default qlen 1000
    link/ether 00:0c:29:4a:af:9c brd ff:ff:ff:ff:ff:ff
    inet 192.168.134.128/24 brd 192.168.134.255 scope global dynamic noprefixroute ens33
       valid_lft 1256sec preferred_lft 1256sec
    inet6 fe80::20c:29ff:fe4a:af9c/64 scope link noprefixroute
       valid_lft forever preferred_lft forever
3: ens33.10@ens33: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc noqueue state UP group default qlen 1000
    link/ether 00:0c:29:4a:af:9c brd ff:ff:ff:ff:ff:ff
    inet 192.168.10.1/24 scope global ens33.10
       valid_lft forever preferred_lft forever
4: ens33.20@ens33: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc noqueue state UP group default qlen 1000
    link/ether 00:0c:29:4a:af:9c brd ff:ff:ff:ff:ff:ff
    inet 192.168.20.1/24 scope global ens33.20
       valid_lft forever preferred_lft forever
maso-prince@maso-prince-VMware-Virtual-Platform:~/CyberOps/Week10$ sudo ip link set ens33 down
maso-prince@maso-prince-VMware-Virtual-Platform:~/CyberOps/Week10$ sudo ip link set ens33 up
maso-prince@maso-prince-VMware-Virtual-Platform:~/CyberOps/Week10$
```

5. Recovery

I planned to restore the network connection and test whether internet access was working again.

sudo ip link set ens33 up

ping -c 4 google.com

```
maso-prince@maso-prince-VMware-Virtual-Platform:~/CyberOps/Week10$ ping -c 4 google.com
PING google.com (142.251.216.78) 56(84) bytes of data.
64 bytes from lcjnbi-ae-in-f14.1e100.net (142.251.216.78): icmp_seq=1 ttl=128 time=11.8 ms
64 bytes from lcjnbi-ae-in-f14.1e100.net (142.251.216.78): icmp_seq=2 ttl=128 time=8.15 ms
64 bytes from lcjnbi-ae-in-f14.1e100.net (142.251.216.78): icmp_seq=3 ttl=128 time=8.83 ms
64 bytes from lcjnbi-ae-in-f14.1e100.net (142.251.216.78): icmp_seq=4 ttl=128 time=7.87 ms

--- google.com ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 2998ms
rtt min/avg/max/mdev = 7.866/9.157/11.777/1.553 ms
maso-prince@maso-prince-VMware-Virtual-Platform:~/CyberOps/Week10$
```

6. Saving evidence

I used the following commands to save system logs, process information and network details.

sudo journalctl -n 100 > incident_logs.txt

ps aux > processes.txt

ip addr > network_status.txt

7. Reflection and analysis

I learned how to identify and investigate a simulated security incident using Ubuntu commands. I used system logs to look for suspicious activity and checked running processes and network information.

This relates to SOC operations because security analysts monitor systems, investigate suspicious events and take action to prevent further damage.

8. Conclusion

I demonstrated the basic incident response process using a simulated security incident. I used system logs and command outputs as evidence during the investigation.

My next step would be to investigate the source of the suspicious activity, contain the threat, remove any malicious activity, restore the system and continue monitoring it.